

CIS Critical Security Controls v8.1 — Compliance Report

Cisco Secure Workload | IG2-Lead Mapping with IG1/IG3 Deltas

Cisco Secure Workload — Compliance Mapping Series

Version 1.0 (draft — pending SME review)

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Cisco Secure Workload — CIS Controls v8.1 Compliance Mapping

How to read this report

This report describes how Cisco Secure Workload (CSW) supports an organisation’s posture against the **CIS Critical Security Controls v8.1** at the workload tier. Default depth is written for **Implementation Group 2 (IG2)**, with IG1 and IG3 deltas called out where the work materially changes.

Why CIS Controls and not just NIST 800-53. The CIS Controls are an operationally ordered, prioritised subset of what NIST 800-53 describes. Where 800-53 lists 1,000+ controls, CIS compresses to **18 Controls** and **~153 Safeguards**, ranked by what defends against actual attacks first. Many programmes use CIS as the “what to do this year” view and 800-53 as the “what to satisfy in the contract” view.

Companion document. The accompanying *CIS Controls v8.1 Technical Runbook* walks through deployment, scope labelling per IG, and the per-Safeguard implementation steps. Read this report for the audit-narrative view; read the runbook for the deployment view.

How to read the Coverage column

Direct means Secure Workload produces the primary evidence artifact for that row — segmentation, policy, flow and process telemetry, inventory, vulnerability and reachability, or forensic reconstruction. Direct is still evidence, not an attestation.

Supporting means Secure Workload contributes an input that feeds or complements another control, tool, or process — detection input only, reconciled against an external register, or one log source among several.

Evidence Required means Secure Workload has no record for that row. The customer supplies the contract, the analysis, or the HR / physical-security evidence from another control.

Every report in this set now uses the same three coverage words — Direct, Supporting, and Evidence Required. Coverage describes what evidence Secure Workload produces; it never means the control is passed.

1. Executive Summary

CIS Controls v8.1 organises 18 Controls and ~153 Safeguards by Implementation Group (IG1 / IG2 / IG3). CSW directly addresses six of the 18 Controls and supports another seven; the remaining five are explicitly outside CSW's scope (training, email/web, backup, and similar non-workload functions).

What CSW changes for a CIS Controls programme:

- **Control 1 / 2 — Inventory.** Continuous, agent-based workload inventory and per-workload software inventory, refreshed at the cadence the IG band expects (monthly for IG1, weekly for IG2, continuous for IG3). Cloud connectors capture ephemeral assets even without an agent.
- **Control 4 — Secure Configuration.** Daily configuration baseline per workload (OS, packages, listening ports, processes) with drift detection and change-ticket attribution.
- **Control 7 — Continuous Vulnerability Management.** Per-workload CVE inventory with CVSS, EPSS, and exploit-availability context, prioritised by reachability in the actual conversation graph (not just CVSS score).
- **Control 8 — Audit Log Management.** Per-flow + per-process audit detail forwarded to your SIEM, retained per the 90-day IG2 minimum (or longer per overlapping framework requirements).
- **Control 13 — Network Monitoring and Defense.** The densest CSW fit. East-west visibility, segmentation enforcement, behavioural detection, conversation-graph anomalies, and SIEM forwarding — the substrate for Safeguards 13.1 through 13.10.
- **Control 17 — Incident Response.** Six-artefact reconstruction bundle for every declared incident, ready to feed playbooks and post-incident review (Safeguard 17.8).

What CSW does not change. CSW does not address Controls 5 (Account Management), 9 (Email & Web Browser), 11 (Data Recovery), 14 (Security Awareness), or the HR/physical sub-Safeguards inside other Controls. Reference your identity, email/web, backup, and training stacks for those.

2. Compliance Posture Summary

Control	Title	Coverage	CSW contribution (IG2 default)
1	Inventory and Control of Enterprise Assets	Direct	Continuous workload + cloud asset inventory; quarantine policy for net-new
2	Inventory and	Direct	Per-workload

Control	Title	Coverage	CSW contribution (IG2 default)
	Control of Software Assets		package inventory; allow-list deviation alerting
3	Data Protection	Supporting	Egress observation (DLP remains primary)
4	Secure Configuration of Enterprise Assets and Software	Direct	Daily baseline + drift report tied to change ticket
5	Account Management	Evidence Required	Identity / PAM tooling
6	Access Control Management	Supporting	Workload-side enforcement; identity is upstream
7	Continuous Vulnerability Management	Direct	Continuous CVE inventory + reachability-aware prioritisation
8	Audit Log Management	Direct	Per-flow + per-process telemetry to SIEM
9	Email and Web Browser Protections	Evidence Required	Email / web stack
10	Malware Defenses	Supporting	Behavioural rules layered with endpoint AV
11	Data Recovery	Evidence Required	Backup product
12	Network Infrastructure Management	Supporting	Workload-side complement to network team config
13	Network Monitoring and Defense	Direct	East-west visibility, segmentation, behavioural detection
14	Security Awareness and Skills Training	Evidence Required	HR / training
15	Service Provider Management	Supporting	Vendor-egress observation (contractual is upstream)
16	Application Software	Supporting	CVE inventory feeds

Control	Title	Coverage	CSW contribution (IG2 default)
	Security		the SDLC
17	Incident Response Management	Direct (evidence)	Six-artefact reconstruction bundle
18	Penetration Testing	Supporting	Pre/post reachability + activity reconstruction

Posture takeaway. Direct on six Controls (1, 2, 4, 7, 8, 13); supporting on seven; out of scope on five. This is consistent with CSW's positioning as a workload-protection platform: it does the workload-tier jobs well and explicitly defers identity, email/web, backup, and training to the appropriate owners.

3. Mapping Table — CIS Safeguard → CSW Capability → Evidence Artefact

Safeguard	Topic (abbreviated)	CSW capability	Evidence artefact (audit-ready)
1.1	Inventory of enterprise assets	Continuous workload inventory + cloud asset metadata	Inventory CSV per scope, dated to IG cadence
1.2	Address unauthorised assets	Quarantine policy applied to net-new workloads	Quarantine log with disposition
1.3, 1.5	Active + passive discovery	Sensor coverage + cloud orchestrator polling	Sensor coverage report + cloud connector status
2.1	Software inventory	Per-workload package + version inventory	Per-workload package inventory dated to IG cadence
2.2	Authorised software supported	Cross-reference vs. vendor-support list	Unsupported-software register
2.3	Address unauthorised software	Alert + (IG3) behavioural-policy block	Unauthorised-software alert log with disposition
2.5–2.6	Allow-listed software / libraries	Allow-list catalogue + behavioural enforcement	Allow-list export + deviation log
4.1	Secure configuration process	Daily baseline + drift report	Baseline export + daily diff log with

Safeguard	Topic (abbreviated)	CSW capability	Evidence artefact (audit-ready)
			change-ticket attribution
4.4	Default deny on FW	CSW segmentation enforcement at workload	Workspace export per scope
4.6	Secure software config	Per-workload listening-port + service inventory	Listening-port inventory dated to IG cadence
6.1–6.5	Access control management	Workload-side identity-aware enforcement	Policy export with rule provenance per scope
7.1–7.5	Continuous vulnerability management	Continuous CVE inventory + reachability prioritisation	CVE list per workload dated to IG cadence
7.7	Remediate detected vulnerabilities	Patch SLA tracking + compensating-control register	Patch SLA compliance trend + compensating-control register
8.2	Collect audit logs	SIEM forwarding configuration	SIEM config screenshot + sample event
8.5	Detailed audit logs	Per-flow + per-process detail	Sample log entries with flow + process attribution
8.7, 8.10	Retain audit logs	SIEM retention + CSW window	Retention policy + retrieval evidence over the window
8.11	Audit log reviews	Quarterly review of CSW alert summary	Review minutes per cadence
10.x	Malware defenses	Behavioural rules layered with endpoint AV	Detection rule catalogue + sample alert
12.2	Secure protocols	Plaintext-flow detection + DENY enforcement	Plaintext-flow alert log + segmentation policy
12.4	Architecture diagrams	ADM dependency export	ADM graph export per scope
13.1, 13.6	Centralised monitoring + flow	CSW flow telemetry + SIEM forwarding	Flow log retention configuration +

Safeguard	Topic (abbreviated)	CSW capability	Evidence artefact (audit-ready)
	logs		sample retrieval
13.2, 13.3	Host + network IDS	Behavioural rules + anomaly detection	Detection rule catalogue + anomaly alert log
13.4	Traffic filtering between segments	CSW segmentation policy enforcement	Workspace export with simulation → enforcement log
13.5	Remote-asset access control	Bastion-source-restricted mgmt protocol allow rules	Allow rule policy export + bastion source-set documentation
13.7–13.10	HIPS, NIPS, port-level, app-layer filtering (IG3)	Behavioural rules + segmentation enforcement	Policy export + behavioural rule catalogue
15.1, 15.4	Service provider management (technical lens)	Vendor-egress observation + reconciliation	Vendor-egress report + reconciliation against contract register
16.1, 16.6	Application security process	CVE inventory feeds SDLC + reachability for triage	CVE feed integration + triage workflow evidence
17.1–17.8	Incident response management	Six-artefact incident dossier	Sample dossier per quarter + MTTR trend
18.1–18.5	Penetration testing	Pre/post reachability + activity reconstruction	Pre/post reachability reports + red-team activity export

4. Out-of-Scope and Complementary Controls

CSW does **not** address the following Controls; reference the noted complementary controls.

Control	Owner	Complementary control
Control 5 — Account Management	Identity / PAM team	AD/LDAP, PAM (CyberArk, BeyondTrust), MFA
Control 9 — Email and Web Browser Protections	Email / web team	Email security gateway, secure web gateway
Control 11 — Data Recovery	Backup / DR team	Backup product + restore testing

Control	Owner	Complementary control
Control 14 — Security Awareness and Skills Training	HR / Security training	LMS + phishing simulation programme
Sub-Safeguards: Physical access, HR, Crypto modules	Facilities / HR / Crypto SME	Physical security, HR process, FIPS 140-validated modules

CSW's role in a complete CIS Controls programme is the workload- tier visibility, segmentation, and continuous evidence engine. It explicitly defers identity, email/web, backup, and training to the appropriate owners.

5. Suggested Next Steps

For organisations adopting or maturing CIS Controls posture with CSW:

1. **Confirm your IG target.** IG1 / IG2 / IG3 calibration drives inventory cadence, scan cadence, log retention, and segmentation depth. Get this signed off by security leadership before scoping.
 2. **Pilot on Controls 1, 2, 7, 13.** These are the densest CSW alignment and produce the fastest visible wins. Inventory and software inventory are the foundation; Controls 7 and 13 are where the daily security work lives.
 3. **Tie drift detection to your change-management ticketing day one.** Orphan changes (Safeguard 4.1) are one of the most asked-for evidence types in CIS assessment, and CSW produces the feed automatically.
 4. **Pair Control 13 with the SIEM you already have.** CSW does not replace the SIEM; it feeds richer, workload-tier data. The uplift to Safeguards 13.1 / 13.3 / 13.6 comes from forwarding CSW telemetry to the SIEM you already operate.
 5. **Build the Control 17 incident dossier template once.** The six-artefact bundle is reusable across every incident; standardise the format so SOC analysts produce identical artefacts under pressure.
 6. **Plan SME review.** This document is a v1 mapping. Before being relied upon in a formal compliance engagement, route to a CIS Controls assessment specialist for currency and to a current Cisco Secure Workload SE for product-capability accuracy.
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6. Disclaimer

This report and its companion runbook describe how Cisco Secure Workload capabilities can support a CIS Critical Security Controls v8.1 programme at the workload tier. They are **not** legal or compliance advice and do not constitute a finding of compliance.

Specifically:

- CIS publishes the authoritative Safeguard text and Implementation Group mapping at <https://www.cisecurity.org/>. Always validate against the **current effective version** of the Controls (v8.1 at the time of authoring; subsequent versions and minor revisions occur).
- Mapping a Safeguard to a CSW capability does not by itself satisfy the Safeguard. Safeguards expect documented process, evidence cadence, and management review — CSW supplies the technical layer that those process requirements depend on.
- CSW does not address Controls 5, 9, 11, 14 or the HR/physical sub-Safeguards within other Controls. Reference your identity, email/web, backup, and training stacks for those.

This document should receive subject-matter-expert review for both CIS Controls accuracy and current Cisco product capability before being used in a customer engagement.